

Section 14.5: The Chain Rule

Partial Derivatives

MTH 310
Calculus III

The Chain Rule — From Calc I to Calc III

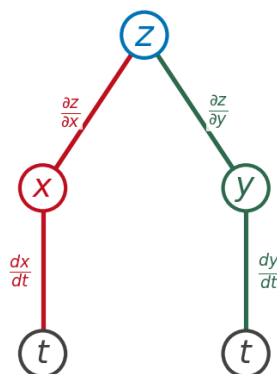
Calc I chain rule: one path, one product. Calc III chain rule: **multiply along each branch, add across branches.**

Case 1: One Independent Variable

Chain Rule (Case 1)

If $z = f(x, y)$ is differentiable, and $x = g(t)$, $y = h(t)$ are differentiable functions of t , then:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$



Draw the tree **first**, then read off the formula. **Multiply along each path, add across paths.**

Example 1: One Independent Variable

Example 1

Let $z = x^2 + y^2 + xy$, where $x = \sin t$ and $y = e^t$.

Find $\frac{dz}{dt}$ at $t = \frac{\pi}{4}$.

Draw the variable tree, write the chain rule formula, compute each piece, then evaluate at $t = \pi/4$.

Your Turn: One Variable Chain Rule

Example 2

Let $z = e^x \sin y$, where $x = t^2$ and $y = 3t$.

Find $\frac{dz}{dt}$.

Draw the variable tree, compute the partials and derivatives, then combine.

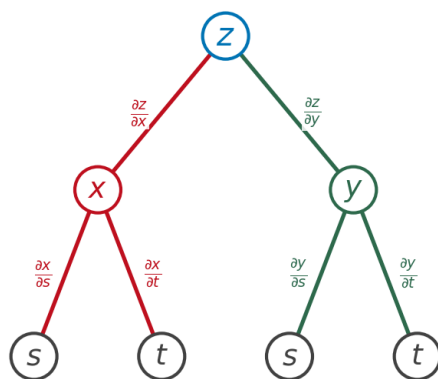
Case 2: Two Independent Variables

Chain Rule (Case 2)

If $z = f(x, y)$, $x = g(s, t)$, $y = h(s, t)$, all differentiable, then:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$



With two independent variables, we get **two formulas** — one for $\partial z / \partial s$ and one for $\partial z / \partial t$. Each follows the same tree pattern.

Example 3: Two Independent Variables

Example 3

Let $z = x^2 y^3$, where $x = s \cos t$ and $y = s \sin t$.

Find $\frac{\partial z}{\partial s}$ when $(s, t) = (2\sqrt{2}, \pi/4)$.

Draw the variable tree for $\partial z/\partial s$. Compute partials, then evaluate at the given point.

The General Chain Rule

General Chain Rule

If $u = f(x_1, x_2, \dots, x_n)$ where each $x_i = g_i(t_1, t_2, \dots, t_m)$, then:

$$\frac{\partial u}{\partial t_j} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_j} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_j} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_j}$$

for each $j = 1, 2, \dots, m$.

You don't need to memorize separate formulas for each case. Just **draw the tree** and read off the answer.

Implicit Differentiation Revisited

Implicit Differentiation Formulas

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

Method 2 is derived **from** the chain rule. It's a convenient shortcut — but you should understand why it works.

Why Does $\partial z/\partial x = -F_x/F_z$ Work?

The formula comes directly from the chain rule applied to $F(x, y, z) = 0$. Same idea gives $\partial z/\partial y = -F_y/F_z$.

Example 4: Implicit Differentiation

Example 4

The equation $x^2 \sin(yz) + 3z = y^2$ defines z implicitly as a function of x and y .

Find $\frac{\partial z}{\partial x}$ using both methods.

Method 1: Differentiate directly w.r.t. x (y constant, $z = z(x,y)$). Method 2: Set $F = 0$, use $\partial z / \partial x = -F_x / F_z$.

Your Turn: Implicit Differentiation

Example 5

The equation $x^3 + y^3 + z^3 + 6xyz = 1$ defines z implicitly as a function of x and y .

Find $\frac{\partial z}{\partial x}$ using the formula $\partial z / \partial x = -F_x / F_z$.

Set $F(x,y,z) = 0$, compute F_x and F_z , then apply the formula.

Homework

Section 14.5: 5, 7, 13, 15, 17, 19, 25, 35, 37, 39, 44, 49